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# Knowing Before Saying: Internal Correctness Signals in Mathematical Reasoning

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## Abstract

1 When a language model is confidently wrong, is the error undetected or left  
2 unsaid? We look inside the model to find out. Across four open models  
3 and four domains (MMLU-Pro, MATH, GPQA-Diamond, and a geometry  
4 task graded by a compiler), a simple linear detector reading the model’s  
5 internal state predicts whether an answer is correct better than the model’s  
6 own stated confidence in 12 of 16 settings, by a mean of +0.09 AUROC.  
7 The gap is widest on MATH (+0.20). The detector is not just spotting  
8 hard questions: given the same question several times, it still separates  
9 the attempts that succeeded from the ones that failed. On Mistral, the  
10 signal exists before we ask for it: on MMLU-Pro, a probe fitted after the  
11 attempt reads at chance before it, and at the last token of the answer, with  
12 no confidence question in play, correctness is readable at 0.69 against 0.57  
13 from the question alone. And on Mistral, though not on GLM, the model’s  
14 report depends on it: remove that direction after the answer is written and  
15 stated confidence can no longer tell the model’s successes from its failures  
16 ( $0.84 \rightarrow 0.56$ ), while removing a random direction of the same size changes  
17 nothing; turn it up and confidence on wrong answers drops while calibration  
18 error halves. Models carry more about their own errors than they report,  
19 and on at least one model, what they report depends on it.

## 20 1 Introduction

21 A language model that is confidently wrong is a problem. Its stated confidence is a signal  
22 for deciding whether to trust its answer. This paper asks whether the model’s activations  
23 contain the signal to know when it might be wrong.

24 Two things could be going on. Upon giving an incorrect answer and saying it has high  
25 confidence, either the model genuinely has no idea it might be wrong and is just blindly  
26 confident, or it deep down holds the error signal in its activations but unsurfaced to its  
27 confidence report. Hence it is either a capability problem or a reporting problem.

28 In this paper we aim to tell these two apart. We read the model’s internal state directly  
29 with a linear detector, a *probe*, and compare what the probe finds to the model’s reported  
30 confidence. We say a fact is *represented* when a probe can read it from the model’s state,  
31 and *reported* when it reaches the confidence number the model writes. The title’s “knowing”  
32 is shorthand for a representation that is present, specific to the attempt, and, in at least one  
33 model, used by the report. We make no claim about awareness, and a probe alone could not  
34 support one.

Prior work has shown that a model’s output probabilities are often better calibrated than its words [4], and that the truth of a given statement can be read from activations [1, 2, 5]. We apply those tools to the model’s own attempts, and then intervene on the representation [6] to ask whether the report depends on it.

## 2 Reading the state

**Models and tasks.** We tested four open models built in four different ways: Mistral-Small-24B (dense), Qwen3.6-27B (gated linear attention interleaved with full attention), and GLM-4.7-Flash and Gemma-4-26B (mixture of experts). Each answered 150 questions, five times each, in four domains: MMLU-Pro, MATH, GPQA-Diamond, and GeoGenBench, a geometry task in which the model writes a construction and a compiler checks it against a formal specification, so its labels need no judge; in a pre-registered check of 200 attempts by two trained raters, the compiler never failed a correct construction. That is 16 cells of about 750 attempts each.

### Prompting structure.

1. The model is shown the question and asked, before answering, how confident it is, from 0 to 100, that it will get it right.
2. It answers.
3. It is asked how confident it is that the answer it just gave is correct.

**The probe.** At the moment the model is about to write its confidence digit, we record the number and the residual-stream hidden state at that token (2,048 to 5,120 dimensions, depending on the model). The probe is a linear classifier on that vector: features are standardized, reduced to 50 principal components, and fed to an L2-regularized logistic regression that predicts whether the attempt was correct. We fit it with five-fold cross-validation grouped by question, so all five attempts at a question fall in the same fold and the probe is always scored on questions it never saw. The read depth is fixed in advance at 70% of the network (layer 28 of 40 for Mistral, 33 of 47 for GLM, 21 of 30 for Gemma, 45 of 64 for Qwen3.6). Every readout is scored with AUROC.

**The activation state carries more than self-report.** The probe beats the model’s stated confidence in **12 of 16** cells, with one tie. The mean gain is +0.09 AUROC (Appendix D). The gap is largest on MATH, at +0.20. Mistral on MATH is a good example of what this looks like. The model is never told whether it got the answer right. Yet after attempting, it raises its confidence, and it raises it more on the attempts it got wrong (+22 points) than on the ones it got right (+16 points). The probe on the same records reads 0.83. The state tracks correctness better than the self-report does.

**Why MATH.** On MATH, right and wrong answers have the same shape. Both are a chain of steps ending in a boxed result, so the model cannot tell from the look of its answer whether it went wrong. Appendix A follows one attempt where a single wrong subtraction leads to a wrong answer, and the model then raises its confidence. Shape still carries a little signal. A baseline classifier that sees only surface features of the answer text, such as its length and line count, reaches 0.74, but the probe reaches 0.83. Geometry is the opposite case. A failed construction usually does not compile, so the failure is visible in the answer itself, and there the surface baseline does as well as the probe. The internal signal matters most where the failure is invisible from outside.

**Controls.** Three things could make a probe look better than it is, and the next section tests each. It could be reading how hard the question is rather than the attempt, so we compare attempts within one question. It could be reading the shape of the output, so we compare it with the surface baseline in every cell; on Mistral the probe’s margin over the baseline is +0.21 on MMLU-Pro, +0.07 on MATH, +0.05 on GPQA, and −0.11 on geometry. And it could be reading something that was there before the attempt, so we check the input layer, where the token is identical in every record and the probe sits at exactly 0.5. For time, these controls were run on Mistral only. Its four cells reproduce the headline.

Table 1: Mistral. Rows: training domain. Columns: test domain. Geo = GeoGenBench.

| train ↓ test → | Geo         | MMLU-Pro    | MATH        | GPQA        |
|----------------|-------------|-------------|-------------|-------------|
| Geo            | <b>0.68</b> | 0.67        | 0.78        | 0.56        |
| MMLU-Pro       | 0.70        | <b>0.75</b> | 0.84        | 0.59        |
| MATH           | 0.80        | 0.72        | <b>0.83</b> | 0.57        |
| GPQA           | 0.60        | 0.69        | 0.74        | <b>0.57</b> |

### 3 What the signal is

**Difficulty control.** A probe that beats stated confidence could be doing something simpler. It could be learning which questions are hard. Hard questions fail more often, so a difficulty detector would predict correctness without reading the attempt. We run two checks.

First, hold the question fixed. Each question is attempted five times, so within one question, difficulty cannot vary. The probe still separates the successful attempts from the failed ones (0.72 to 0.74 in representative cells). The model’s own  $P(\text{True})$ , its probability of answering “True” when asked whether it was right, drops to near chance (0.47 to 0.59).

Second, test where the direction lives. Before the attempt, at the last token of the question, the model’s state is the same across all five attempts. It can encode difficulty and nothing else. A probe trained after the attempt and read at that point scores 0.49 on MMLU-Pro, which is chance, against 0.75 at its own site. The direction that separates good attempts from bad ones does not exist until the model has tried. Geometry is the exception. There the pre-attempt state already predicts failure, so that cell is mostly difficulty.

**Reading before the question.** There is a subtler worry. Every read so far came from a token the model reaches after being asked to assess itself. Maybe the question creates the signal rather than reveals it. So we also read the state at the last token of the answer, before any confidence question exists. On MMLU-Pro, correctness is already readable there at 0.69. From the question alone it is 0.57, so the model cannot see failure coming. Asking the model to self-assess sharpens the signal to 0.77.

**Transfer across domains.** Table 1 trains a probe on one domain and tests it on each of the others. On Mistral it keeps about 90% of its lift above chance (0.69 off the diagonal against 0.71 on it). The original matrix showed the same pattern in every model. The one exception is Gemma-4. There, probes trained on geometry fail elsewhere (0.31 to 0.39), while probes trained elsewhere work on geometry (about 0.70). A probe trained on the one generative task can learn features specific to it.

### 4 Whether self-report depends on the residual signal

**Intervention.** A model could carry a perfect internal record of its errors and never consult it during self-report. The only way to find out is to change the record and watch the number. At the confidence token, after the answer is written so the answer cannot change, we scale the model’s position along the correctness direction by a gain  $g$ . Gain 1 leaves the model untouched, gain 0 erases the per-attempt information, and gains 2 and 4 exaggerate it. We run this on Mistral and MATH, 150 held-out records, with a random direction of the same size as a control. Table 2 shows what the model wrote at each setting.

**Result on Mistral.** Erase the direction and self-report loses its information: AUROC falls from 0.84 to 0.56, a change of  $-0.27$   $[-0.42, -0.12]$ . Erase a random direction and nothing happens  $(-0.00 [-0.03, +0.03])$ . This is not generic damage. The output stays well formed, and confidence on wrong answers goes *up* to match right answers, so the model becomes uniformly sure rather than confused. Exaggerate the direction and confidence on wrong answers falls while right answers hold; calibration error halves,  $0.34 \rightarrow 0.19$ . Two checks say the effect is specific. The same intervention 30% of the way through the network

Table 2: Mistral on MATH, 150 held-out records. The correctness direction scaled by a gain at the confidence token, after the answer is written.

| gain on the correctness direction      | untouched<br>1 | erased<br>0    | doubled<br>2 | quadrupled<br>4 |
|----------------------------------------|----------------|----------------|--------------|-----------------|
| mean confidence, wrong / right answers | 84 / 96        | <b>97 / 98</b> | 70 / 94      | <b>54 / 88</b>  |
| AUROC of stated confidence             | 0.84           | <b>0.56</b>    | 0.85         | 0.82            |
| same, with a random direction          | 0.84           | 0.83           | 0.84         | output broke    |

127 does nothing, and a direction learned on MMLU-Pro, where surface features cannot explain  
128 the probe, still controls the MATH report ( $0.88 \rightarrow 0.71$ ).

129 **Result on GLM.** GLM has the strongest signal in the study: within a question, the  
130 probe reads 0.87 while the model’s own  $P(\text{True})$  reads 0.53. Yet erasing the direction leaves  
131 self-report where it was, 0.76 before and after. GLM represents its errors more sharply than  
132 Mistral and reports them less. This is our cleanest case of represented but not reported, and  
133 it makes the causal result a fact about particular models, not models in general.

## 134 5 The Jacobian lens

135 **A label-free check.** A trained probe might have overfit to our data. So we added an  
136 instrument that uses no correctness labels. The Jacobian lens [3] is fit on generic text and  
137 reads an activation for what it pushes the model to say next; we score each state by how  
138 far it leans toward “wrong” over “correct.” On dense models the lens matches the probe:  
139 on Mistral it reads 0.82 on MATH and 0.75 on MMLU-Pro, against the probe’s 0.83 and  
140 0.75, with a clean input-layer control. Two instruments built from different evidence find the  
141 same direction, and because the lens reads through the model’s own output pathway, that  
142 direction sits where it could influence what the model says.

143 **Where the lens fails.** The lens is a fixed dictionary from hidden-state directions to words,  
144 built once from generic text. On a dense model one dictionary is enough, because every  
145 input takes the same path to the output. GLM routes each input through a different subset  
146 of experts, and Qwen3.6 mixes two kinds of attention layers, so there is no single path and  
147 the lens averages many into a blur. The numbers show it: on GLM the lens’s entries for  
148 “correct” and “wrong” point almost the same way, cosine 0.96, against 0.4 on a dense model,  
149 and it scores 0.31 there and 0.43 on Qwen3.6, while the probe still works on both. Swap  
150 Qwen3.6 for Qwen2.5-14B, same family but dense, and the lens jumps to 0.88. The signal is  
151 there in every model. A label-free reader only works where the architecture gives it one path  
152 to read.

## 153 6 What this means

154 **Practical use.** Without an exact checker, the probe is the best per-attempt readout we  
155 found. Keeping the attempts it scores highest raises accuracy far more than keeping the  
156 ones the model is most confident in, and it works where the model’s confidence is useless.

157 **Limitations.** A probe could be reading properties of failed output rather than a self-  
158 assessment; the surface baseline bounds this without closing it, most loosely on MATH,  
159 geometry (Appendix C), and GLM. Every number is a single seed. The before-asking test  
160 runs only on MMLU-Pro; on MATH the question alone already predicts correctness at 0.81.  
161 The causal result holds on Mistral and fails on GLM; Gemma-4 leaves too few failures to  
162 test, and the Qwen3.6 run did not finish in time. Whether preference tuning creates the gap  
163 is open; a base model would answer it.

164 **Takeaway.** These models carry information about their own errors that their self-report  
165 does not, and on one model self-report depends on it.

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## A The elicitation protocol, and one record end to end

This appendix gives the exact prompts behind every record, then one complete record from the Mistral-Small-24B  $\times$  MATH cell, summarized first and then given in full.

**One record, Mistral-Small-24B  $\times$  MATH.** *What integer  $n$  satisfies  $0 \leq n < 18$  and  $n \equiv -11213141 \pmod{18}$ ? The model sets the problem up correctly and gets  $18 \times 622953 = 11213154$  right, then writes  $r = -11213141 + 11213154 = 3$  and answers 3. That subtraction gives 13, which is the gold answer. Stated confidence: 90 before the attempt, 95 after it. Other readouts of the same attempt: P(True), the probability the model assigns to “True” when asked whether its answer is correct, 0.32; mean answer log-prob  $-0.05$ .*

A single arithmetic slip inside a fluent derivation. Having just made the error, the model revises its stated confidence upward, while P(True) on the same transcript assigns only 0.32 probability to correctness. The rest of this appendix gives the prompts and the full transcript.

### A.1 Prompts

Each record is four forward passes: the three turns of Section 2 plus one read taken beforehand, at the last prompt token of the bare question, before any elicitation. The confidence turns run with thinking off so the read lands on a clean one-line reply; the attempt turn runs with thinking on for hybrid reasoners. The fixed **Confidence**: N format is what makes the read site content-neutral: the digit token has identical local context on every record, so layer-0 activations carry no signal by construction.

#### System prompt (MATH).

Solve the problem step by step. Put your final answer in `\boxed{}`.

#### Turn 1, prospective confidence. The problem is followed by:

Before answering, estimate how likely you are to answer this question correctly. Reply with EXACTLY one line and nothing else:

208 Confidence: N  
 209 where N is an integer from 0 (certain to be wrong) to 100 (certain to be  
 210 correct). Do NOT answer yet.

211 **Turn 2, the attempt.** The turn-1 reply is left in context as the assistant message, then:

212 Now give your final answer inside `\boxed{}`.

213 **Turn 3, retrospective confidence.** The attempt is left in context, then:

214 Now assess the answer you just gave. On a scale of 0 to 100, how confident  
 215 are you that it is correct? Reply with exactly one line and nothing else:  
 216 Confidence: N

217 **P(True) baseline.** A separate teacher-forced branch off turn 2, no generation. Next-token  
 218 mass on True versus False:

219 Consider your final answer above. Is it correct? Reply with exactly one word:  
 220 True or False.

221 Grading is external and is never shown to the model, so the turn-3 revision is blind self-  
 222 assessment.

223 **A.2 One record, verbatim**

224 Record `math_test_number_theory_753`, sample 2 of 5.

225 **Problem.** What integer  $n$  satisfies  $0 \leq n < 18$  and  $n \equiv -11213141 \pmod{18}$ ?

226 **Turn 1 reply.** Confidence: 90

227 **Turn 2 reply (complete).**

228 First, we need to find the remainder when  $-11213141$  is divided by 18.  
 229 1. Calculate  $-11213141 \bmod 18$ :

$$-11213141 \div 18 \approx -622952.2778$$

230 The integer part of the division is  $-622953$ .

$$-11213141 = 18 \times (-622953) + r$$

231 where  $r$  is the remainder.  
 232 2. Calculate the remainder  $r$ :

$$r = -11213141 - 18 \times (-622953)$$

$$r = -11213141 + 11213154$$

$$r = 3$$

235 Therefore,

$$-11213141 \equiv 3 \pmod{18}$$

236 So, the integer  $n$  that satisfies  $0 \leq n < 18$  and  $n \equiv -11213141 \pmod{18}$  is 3.

237 **Turn 3 reply.** Confidence: 95

238 **What happened.** The method is right and every intermediate is right:  $18 \times 622953 =$   
 239  $11213154$  checks out, and the model has written the correct subtraction on the line above its  
 240 answer. It then evaluates  $-11213141 + 11213154$  as 3 rather than 13. Gold is 13. Nothing  
 241 in the presentation marks the slip. The derivation is well-formed, correctly structured, and  
 242 confidently laid out, which is precisely why stated confidence rises from 90 to 95 across an  
 243 attempt that failed.

244 **Readouts of this attempt.**

|     | readout                                   | value            |
|-----|-------------------------------------------|------------------|
|     | stated confidence, prospective (turn 1)   | 90               |
|     | stated confidence, retrospective (turn 3) | 95               |
| 245 | P(True)                                   | 0.32             |
|     | mean answer log-probability               | -0.05            |
|     | graded outcome                            | <b>incorrect</b> |

246 This is one record, chosen to illustrate the failure mode rather than to carry evidential weight;  
 247 representative cell-level numbers are in Table 3, Appendix D.

## 248 **B Calibration beyond ECE, and cell-level multiplicity**

249 ECE depends on binning and on sample composition, so we report two binning-free proper  
 250 scoring rules alongside it. On Mistral  $\times$  MATH (750 records, accuracy 0.648), the model’s  
 251 retrospective stated confidence gives Brier 0.262 and NLL 1.470 against a 10-bin ECE of  
 252 0.271. P(True) on the same records gives Brier 0.386: better ordered than stated confidence,  
 253 worse calibrated in absolute terms. The steering results in Table 2 are reported as ECE  
 254 because the steered runs stored per-gain aggregates rather than per-record confidences;  
 255 recomputing Brier and NLL at each gain requires re-running the intervention and is the first  
 256 thing we would add.

257 The cell-level bootstrap counts in Table 3 are uncorrected. With 16 cells, some individually  
 258 significant results are expected under the null, so the cell counts should be read as descriptive.  
 259 The aggregate pattern is the claim that carries weight: the effect is positive in 12 of 16 cells  
 260 with one tie, significantly positive in 10, and never significantly negative. A hierarchical  
 261 model pooling across cells with model and domain as random effects is the right test and is  
 262 not yet run.

## 263 **C Alternative explanations for the probe signal**

264 A probe could key on properties of failed derivations rather than on a dedicated metacognitive  
 265 variable: low token probability, hedging language, or, on MATH, an early-layer channel next  
 266 to the answer text. Our surface-features baseline (answer length, digit count, line count,  
 267 word count, brace count) bounds this without closing it, and the bound is domain-dependent:  
 268 the probe adds +0.21 over surface on MMLU-Pro but only +0.07 on MATH and -0.11  
 269 on geometry, where a construction that fails to compile is visibly malformed. Two results  
 270 argue against a pure surface account on the domains where the signal is strong. A direction  
 271 fitted on MMLU-Pro, where surface reaches only 0.54 against a probe of 0.75, still governs  
 272 MATH’s self-report under ablation (0.880  $\rightarrow$  0.709). And the pre-attempt read, taken where  
 273 the activation is byte-identical across a question’s attempts, cannot recover the post-attempt  
 274 direction on MMLU-Pro (0.490, chance), which a surface-of-the-question account would  
 275 predict it could. A linear probe at one token also undercounts nonlinear or multi-token  
 276 signal, so the reported numbers are a floor.

## 277 **D Per-cell numbers and specificity controls**

## 278 **E Checkpoints**

279 All models are public Hugging Face releases, run in bfloat16 without quantization,  
 280 greedy decoding: mistralai/Mistral-Small-24B-Instruct-2501, Qwen/Qwen3.6-27B,  
 281 zai-org/GLM-4.7-Flash, google/gemma-4-26B-A4B-it; the dense within-family control  
 282 for the lens is Qwen/Qwen2.5-14B-Instruct.

Table 3: Post-attempt correctness discrimination (AUROC). Left: stated confidence vs. the probe in representative cells; overall the probe wins 12 of 16 with one tie, mean +0.09, with a paired bootstrap significantly positive in 10 of 16 and never significantly negative, uncorrected at the cell level (Appendix B). Right: Mistral specificity controls. “pre” is the last prompt token before the attempt; “post” is after the attempt; “resid.” removes the pre score; “post→pre” applies that fitted post readout at the pre-attempt site, where only difficulty is visible.

| <b>model × domain</b> | <b>stated</b> | <b>probe</b> | <b>Mistral</b> | <b>pre</b> | <b>post</b> | <b>resid.</b> | <b>post→pre</b> |
|-----------------------|---------------|--------------|----------------|------------|-------------|---------------|-----------------|
| Gemma-4 × MATH        | 0.57          | <b>0.78</b>  | MMLU-Pro       | 0.568      | 0.754       | 0.751         | 0.490           |
| Qwen3.6 × MMLU-Pro    | 0.67          | <b>0.84</b>  | MATH           | 0.792      | 0.834       | 0.701         | 0.624           |
| GLM × MMLU-Pro        | 0.67          | <b>0.85</b>  | GPQA           | 0.593      | 0.572       | 0.544         | 0.479           |
| Mistral × MATH        | 0.79          | <b>0.83</b>  | GeoGen         | 0.770      | 0.676       | 0.505         | 0.648           |